WATER WORLD
Viewed from space, what clearly makes Earth unique is the abundance of surface water—in the oceans, lakes, atmosphere, and polar ice caps. The presence of surface water has been a key factor in the development of life on Earth.

EARTH PROFILE

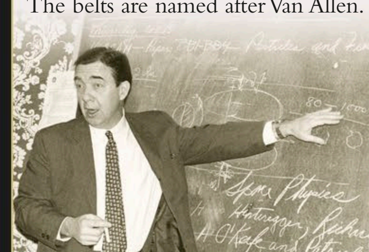
AVERAGE DISTANCE FROM THE SUN 93.0 million miles (149.6 million km)	**ROTATION PERIOD** 23.93 hours
AVERAGE SURFACE TEMPERATURE 59°F (15°C)	**ORBITAL PERIOD (LENGTH OF YEAR)** 365.26 days
DIAMETER 7,926 miles (12,756 km)	**MASS (EARTH = 1)** 1
VOLUME (EARTH = 1) 1	**GRAVITY AT EQUATOR (EARTH = 1)** 1
NUMBER OF MOONS 1	

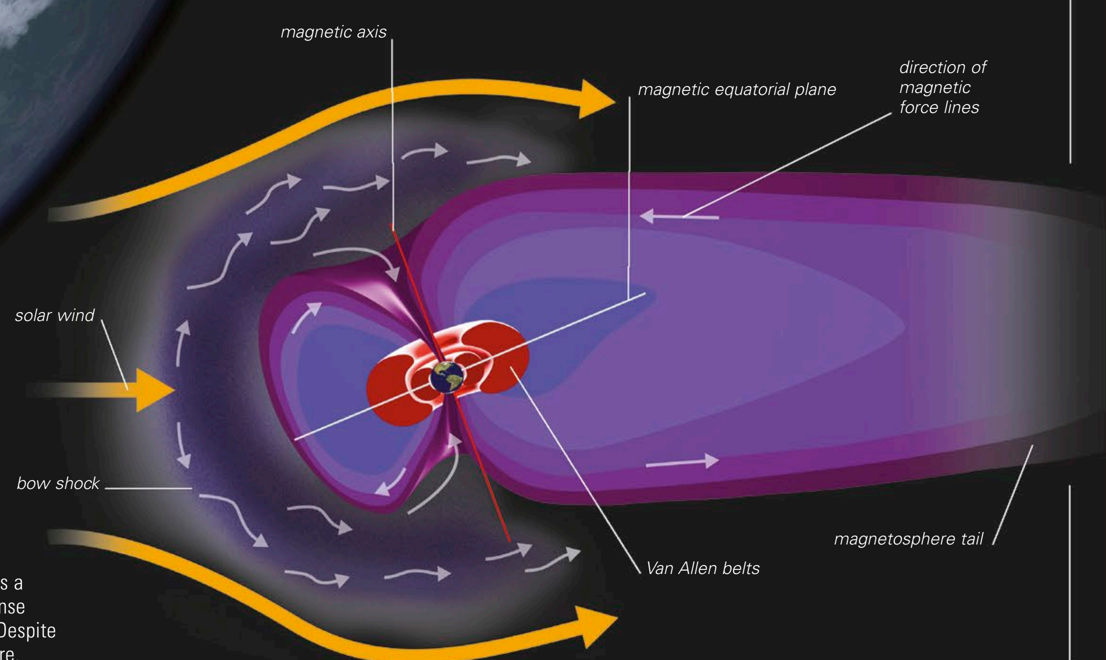
MAGNETIC FIELD

Earth has a substantial magnetic field, which is thought to be caused by a swirling motion of its liquid metal outer core. This motion is driven by a combination of Earth's rotation and convection currents within the outer core. The magnetic field behaves as though a large bar magnet were present within the Earth, tilted at an angle to its axis of rotation. The lines of the magnetic field converge at two points on Earth's surface called the north and south magnetic poles. The location of these points slowly changes over time. Currently, the north magnetic pole is north of Canada in the Arctic Ocean, while the south magnetic pole is north of eastern Antarctica, in the Southern Ocean. The magnetic field extends into space, forming a protective blanket around the planet by deflecting high-speed streams of charged particles that flow toward Earth in the solar wind (see p.107). A few of the particles escape deflection and become trapped within two regions surrounding Earth called the Van Allen radiation belts (see panel, right). Studies of iron-rich minerals in Earth's crust have shown that at variable time intervals (from less than 100,000 to millions of years), Earth's north and south magnetic poles switch.

JAMES VAN ALLEN

James Van Allen (1914–2006) is an American physicist who, in the 1950s, designed and built instruments for American satellites. In 1958, a Van Allen-designed instrument carried by the first US satellite, Explorer 1, detected two large, doughnut-shaped belts of radiation around Earth, which carry trapped charged particles. The belts are named after Van Allen.





EARTH'S MAGNETOSPHERE
The imaginary surface at which Earth's magnetic field first deflects the solar wind is called the bow shock. Behind it is a region of space dominated by the magnetic field, in the sense that the field prevents solar wind particles from entering. Despite its elongated shape, this region is called the magnetosphere.